

1. What's the output of the following code ?

```
Java
1 int doSomething(int a, int b)
2 {
3     if (b==1)
4         return a;
5     else
6         return a + doSomething(a,b-1);
7 }
8
9     doSomething(2,3);
```

- a)4
- b)2
- c)3
- d)6

Ans d

2. Select the correct output.

```
Java
1 int rec(int num){
2     return (num) ? num%10 +
3     rec(num/10):0;
4 }
5
6     main(){
7
```

```
8 printf("%d",rec(4567));  
   }
```

- a) 4
- b) 12
- c) 22
- d) 21

Ans. c

3. Choose correct option.

```
Java  
1 int something(int number)  
2 {  
3     if(number <= 0)  
4         return 1;  
5     else  
6         return number * something(number-  
7 1);  
8  
9 }  
10  
something(4);
```

- a) 12
- b) 24
- c) 1
- d) 0

Ans.b

4. What will be the output

```
int func(int a, int b){
```

```
    if(b==0)
```

```
        return 0;
```

```
    if(b==1)
```

```
        return a;
```

```
    return a + func(a,b-1);
```

```
}
```

what will be the output of **func(3,8)** .

- a) 11
- b) 24
- c) 22
- d) 21

Ans.b

5. What will be the output

```
void print(int n)
```

```
{
```

```
    if (n == 0)
```

```
        return;
```

```
    printf("%d", n%2);
```

```
print(n/2);  
}
```

What will be the output of **print(12)**.

- a) 0011
- b) 1100
- c) 1001
- d) 1000

Ans.a

6. What will be the output

```
int sum(int n) {  
  
    if (n==0)  
        return n;  
    else  
        return n + sum(n-1);  
}
```

What will be the output of **sum(8)**.

- a) 40
- b) 36
- c) 8
- d) 15

Ans.b

7. What will be the output of the following C code?

```
#include<stdio.h>
main()
{
    int n;
    n=f1(4);
    printf("%d",n);
}
f1(int x)
{
    int b;
    if(x==1)
        return 1;
    else
        b=x*f1(x-1);
    return b;
}
```

- a) 24
- b) 4
- c) 12
- d) 10

Ans a

8. What will be the output of the following C code?

```
#include<stdio.h>
main()
{
    int n,i;
    n=f(6);
    printf("%d",n);
}
f(int x)
{
    if(x==2)
        return 2;
```

```
else
{
    printf("+");
    f(x-1);
}
}
```

- a) ++++2
- b) +++++2
- c) ++++++
- d) 2

Ans. A

9. How many times is 'a' printed when the following C code is executed?

```
#include<stdio.h>
main()
{
    int a;
    a=f1(10);
    printf("%d",a);
}
f1(int b)
{
    if(b==0)
        return 0;
    else
    {
        printf("a");
        f1(b--);
    }
}
```

- a) 9 times
- b) 10 times
- c) 0 times
- d) Infinite number of times

Ans.d

10. What will be the output of the following C code?

```
#include<stdio.h>
main()
{
    int n=10;
    int f(int n);
    printf("%d",f(n));
}
int f(int n)
{
    if(n>0)
        return(n+f(n-2));
}
```

- a) 10
- b) 80
- c) 30
- d) Error

Ans. c

11. What will be the output of the following C code?

```
#include<stdio.h>
int main()
{
    printf("Hello");
    main();
}
```

```
    return 0;  
}
```

- a) Hello is printed once
- b) Hello infinite number of times
- c) Hello is not printed at all
- d) 0 is returned

Ans.b

12. What will be the output of the following C code if the input given to the code shown below is "skillbout"?

```
#include<stdio.h>  
#define NL '\n'  
main()  
{  
    void f(void);  
    printf("enter the word\n");  
    f();  
}  
void f(void)  
{  
    char c;  
    if((c=getchar())!=NL)  
    {  
        f();  
        printf("%c",c);  
    }  
    return;  
}
```

- a) skillbout
- b) infinite loop
- c) tuoblliks
- d) likstuobl

Ans. c

13. Predict output of following program

```
#include <stdio.h>
```

```
int fun(int n)
{
    if (n == 4)
        return n;
    else return 2*fun(n+1);
}
```

```
int main()
{
    printf("%d ", fun(2));
    return 0;
}
```

- a. 4
- b. 8
- c. 16
- d. Runtime error

Ans. C

14. Consider the following recursive function fun(x, y). What is the value of fun(4, 3)

```
int fun(int x, int y)
{
    if (x == 0)
        return y;
    return fun(x - 1, x + y);
}
```

- a. 13
- b. 12
- c. 9
- d. 10

Ans. A

15. What does the following function do?

```
int fun(int x, int y)
{
    if (y == 0)    return 0;
    return (x + fun(x, y-1));
}
```

- a. x+y
- b. x+x*y
- c. x*y
- d. x^y

Ans c

16. Output of following program?

```
#include<stdio.h>

void print(int n)
{
    if (n > 4000)
        return;
    printf("%d ", n);
    print(2*n);
    printf("%d ", n);
}

int main()
{
    print(1000);
    getchar();
    return 0;
}
```

- a. 1000 2000 4000
- b. 1000 2000 4000 4000 2000 1000
- c. 1000 2000 4000 2000 1000
- d. 1000 2000 2000 1000

Ans. B

17.What does the following function do?

```
int fun(unsigned int n)
{
    if (n == 0 || n == 1)
        return n;

    if (n%3 != 0)
        return 0;

    return fun(n/3);
}
```

- a. It returns 1 when n is a multiple of 3, otherwise returns 0
- b. It returns 1 when n is a power of 3, otherwise returns 0
- c. It returns 0 when n is a multiple of 3, otherwise returns 1
- d. It returns 0 when n is a power of 3, otherwise returns 1

Ans. B

18.What will be the Output of following program?

```
#include <stdio.h>
int fun(int n, int *f_p)
{
    int t, f;
    if (n <= 1)
```

```

    {
        *f_p = 1;
        return 1;
    }
    t = fun(n- 1,f_p);
    f = t+ * f_p;
    *f_p = t;
    return f;
}

int main()
{
    int x = 15;
    printf (" %d n", fun(5, &x));
    return 0;
}

```

- a. 6
- b. 8
- c. 14
- d. 15

Ans. B

19.Consider the following C function:

```
int f(int n)
{
    static int i = 1;
    if (n >= 5)
        return n;
    n = n+i;
    i++;
    return f(n);
}
```

The value return by f(1) is

- a. 5
- b. 6
- c. 7
- d. 8

Ans c

20. What will be the output

```
void my_recursive_function()
{
    my_recursive_function();
}
int main()
{
    my_recursive_function();
    return 0;
}
```

- a) The code will be executed successfully and no output will be generated
- b) The code will be executed successfully and random output will be

generated

c) The code will show a compile time error

d) The code will run for some time and stop when the stack overflows

Ans .d

21. What is the output of the following code?

```
void my_recursive_function(int n)
{
    if(n == 0)
        return;
    printf("%d ",n);
    my_recursive_function(n-1);
}
int main()
{
    my_recursive_function(10);
    return 0;
}
```

a) 10

b) 1

c) 10 9 8 ... 1 0

d) 10 9 8 ... 1

Ans. D

22. What is the base case for the following code?

```
void my_recursive_function(int n)
```

```
{
    if(n == 0)
        return;
    printf("%d ",n);
    my_recursive_function(n-1);
}
int main()
{
    my_recursive_function(10);
    return 0;
}
```

- a) return
- b) printf("%d ", n)
- c) if(n == 0)
- d) my_recursive_function(n-1)

Ans. C

23. How many times is the recursive function called, when the following code is executed?

```
void my_recursive_function(int n)
{
    if(n == 0)
        return;
    printf("%d ",n);
    my_recursive_function(n-1);
}
int main()
{
    my_recursive_function(10);
    return 0;
}
```


- a) 9
- b) 10
- c) 11
- d) 12

Ans. C

25. What does the following recursive code do?

```
void my_recursive_function(int n)
{
    if(n == 0)
        return;
    my_recursive_function(n-1);
    printf("%d ",n);
}
int main()
{
    my_recursive_function(10);
    return 0;
}
```

- a) Prints the numbers from 10 to 1
- b) Prints the numbers from 10 to 0
- c) Prints the numbers from 1 to 10
- d) Prints the numbers from 0 to 10

Ans c

26. will be the output of the following code?

```
int cnt=0;
void my_recursive_function(int n)
{
```

```
    if(n == 0)
        return;
    cnt++;
    my_recursive_function(n/10);
}
int main()
{
    my_recursive_function(123456789);
    printf("%d",cnt);
    return 0;
}
```

- a) 123456789
- b) 10
- c) 0
- d) 9

Ans d

27. What will be the output of the following code?

```
void my_recursive_function(int n)
{
    if(n == 0)
    {
        printf("False");
        return;
    }
    if(n == 1)
    {
        printf("True");
        return;
    }
    if(n%2==0)
        my_recursive_function(n/2);
    else
    {
```

```

        printf("False");
        return;
    }
}
int main()
{
    my_recursive_function(100);
    return 0;
}

```

- a) True
- b) False

Ans. B

28. What is the output of the following code?

```

int cnt = 0;
void my_recursive_function(char *s, int i)
{
    if(s[i] == '\0')
        return;
    if(s[i] == 'a' || s[i] == 'e' || s[i] == 'i' || s[i] ==
'o' || s[i] == 'u')
        cnt++;
    my_recursive_function(s,i+1);
}
int main()
{
    my_recursive_function("thisisrecursion",0);
    printf("%d",cnt);
    return 0;
}

```

- a) 6
- b) 9

- c) 5
- d) 10

Ans. a

29. What is the output of the following code?

```
void my_recursive_function(int *arr, int val, int idx, int len)
{
    if(idx == len)
    {
        printf("-1");
        return ;
    }
    if(arr[idx] == val)
    {
        printf("%d",idx);
        return;
    }
    my_recursive_function(arr,val,idx+1,len);
}
int main()
{
    int array[10] = {7, 6, 4, 3, 2, 1, 9, 5, 0, 8};
    int value = 2;
    int len = 10;
    my_recursive_function(array, value, 0, len);
    return 0;
}
```

- a) 3
- b) 4
- c) 5
- d) 6

Ans. b

30. Consider the following iterative implementation to find the factorial of a number. Which of the lines should be inserted to complete the below code?

```
int main()
{
    int n = 6, i;
    int fact = 1;
    for(i=1;i<=n;i++)
        _____;
    printf("%d",fact);
    return 0;
}
```

- a) fact = fact + i
- b) fact = fact * i
- c) i = i * fact
- d) i = i + fact

And. B

31. Consider the following recursive implementation to find the factorial of a number. Which of the lines should be inserted to complete the below code?

```
int fact(int n)
{
    if(_____)
        return 1;
    return n * fact(n - 1);
}
int main()
{
    int n = 5;
```

```
    int ans = fact(n);  
    printf("%d",ans);  
    return 0;  
}
```

- a) $n = 0$
- b) $n \neq 0$
- c) $n == 0$
- d) $n == 1$

Ans. C

32.The time complexity of the following recursive implementation to find the factorial of a number is _____

```
int fact(int n)  
{  
    if(_____)  
        return 1;  
    return n * fact(n - 1);  
}  
int main()  
{  
    int n = 5;  
    int ans = fact(n);  
    printf("%d",ans);  
    return 0;  
}
```

- a) $O(1)$
- b) $O(n)$
- c) $O(n^2)$
- d) $O(n^3)$

Ans. B

33. Consider the following recursive implementation to find the factorial of a number. Which of the lines is the base case?

```
int fact(int n)
{
    if(n == 0)
        return 1;
    return n * fact(n - 1);
}
int main()
{
    int n = 5;
    int ans = fact(n);
    printf("%d",ans);
    return 0;
}
```

- a) return 1
- b) return n * fact(n-1)
- c) if(n == 0)
- d) if(n == 1)

Ans. C

34. What is the output of the following code?

```
int fact(int n)
{
    if(n == 0)
        return 1;
    return n * fact(n - 1);
}
int main()
{
    int n = 0;
    int ans = fact(n);
    printf("%d",ans);
}
```

```
    return 0;
}
```

- a) 0
- b) 1
- c) 2
- d) 3

Ans. B

35. What is the output of the following code?

```
int fact(int n)
{
    if(n == 0)
        return 1;
    return n * fact(n - 1);
}
int main()
{
    int n = 1;
    int ans = fact(n);
    printf("%d",ans);
    return 0;
}
```

- a) 0
- b) 1
- c) 2
- d) 3

Ans b

36. What is the output of the following code?

```
int fact(int n)
{
    if(n == 0)
        return 1;
    return n * fact(n - 1);
}
int main()
{
    int n = 5;
    int ans = fact(n);
    printf("%d",ans);
    return 0;
}
```

- a) 24
- b) 120
- c) 720
- d) 1

Ans. B

37. Consider the following iterative solution to find the sum of first n natural numbers:

```
#include<stdio.h>
int get_sum(int n)
{
    int sm = 0, i;
    for(i = 1; i <= n; i++)
        _____;
    return sm;
}
int main()
{
    int n = 10;
```

```
    int ans = get_sum(n);  
    printf("%d",ans);  
    return 0;  
}
```

Which of the following lines completes the above code?

- a) sm = i
- b) sm += i
- c) i = sm
- d) i += sm

Ans. B

38. What is the output of the following code?

```
#include<stdio.h>  
int get_sum(int n)  
{  
    int sm, i;  
    for(i = 1; i <= n; i++)  
        sm += i;  
    return sm;  
}  
int main()  
{  
    int n = 10;  
    int ans = get_sum(n);  
    printf("%d",ans);  
    return 0;  
}
```

- a) 55
- b) 45
- c) 35
- d) Depends on compiler

Ans d

39. What is the output of the following code?

```
#include<stdio.h>
int recursive_sum(int n)
{
    if(n == 0)
        return 0;
    return n + recursive_sum(n - 1);
}
int main()
{
    int n = -4;
    int ans = recursive_sum(n);
    printf("%d",ans);
    return 0;
}
```

- a) 0
- b) -10
- c) 1
- d) runtime error

Ans. D

40. Consider the following iterative implementation to find the sum of digits of a number:

```
#include<stdio.h>
int sum_of_digits(int n)
{
    int sm = 0;
    while(n != 0)
    {
```

```

        }
        n /= 10;
    }
    return sm;
}
int main()
{
    int n = 1234;
    int ans = sum_of_digits(n);
    printf("%d", ans);
    return 0;
}

```

Which of the following lines should be inserted to complete the above code?

- a) sm += n
- b) sm += n%10
- c) sm += n-10
- d) sm += n/10

Ans b

41. What is the output of the following code?

```

#include<stdio.h>
int cnt =0;
int my_function(int n, int sm)
{
    int i, tmp_sm;
    for(i=1;i<=n;i++)
    {
        tmp_sm = recursive_sum_of_digits(i);
        if(tmp_sm == sm)
            cnt++;
    }
    return cnt;
}

```

```

}
int recursive_sum_of_digits(int n)
{
    if(n == 0)
        return 0;
    return n % 10 + recursive_sum_of_digits(n/10);
}
int main()
{
    int n = 20, sum = 3;
    int ans = my_function(n,sum);
    printf("%d",ans);
    return 0;
}

```

- a) 0
- b) 1
- c) 2
- d) 3

42. Consider the following iterative implementation used to reverse a string:

```

#include<stdio.h>
#include<string.h>
void reverse_string(char *s)
{
    int len = strlen(s);
    int i,j;
    i=0;
    j=len-1;
    while(_____)
    {
        char tmp = s[i];
        s[i] = s[j];
        s[j] = tmp;
    }
}

```

```
        i++;  
        j--;  
    }  
}
```

Which of the following lines should be inserted to complete the above code?

- a) $i > j$
- b) $i < \text{len}$
- c) $j > 0$
- d) $i < j$

Ans d

43. is the output of the following code?

```
#include<stdio.h>  
#include<string.h>  
void reverse_string(char *s)  
{  
    int len = strlen(s);  
    int i,j;  
    i=0;  
    j=len-1;  
    while(i < j)  
    {  
        char tmp = s[i];  
        s[i] = s[j];  
        s[j] = tmp;  
        i++;  
        j--;  
    }  
}  
int main()  
{  
    char s[100] = "reverse";
```

```
reverse_string(s);  
printf("%s",s);  
return 0;  
}
```

- a) ersevre
- b) esrever
- c) eserver
- d) eresevr

Ans b

44.What does the following code do?

```
#include<stdio.h>  
#include<string.h>  
void reverse_string(char *s)  
{  
    int len = strlen(s);  
    int i,j;  
    i=0;  
    j=len-1;  
    while(i < j)  
    {  
        char tmp = s[i];  
        s[i] = s[j];  
        s[j] = tmp;  
        i++;  
        j--;  
    }  
}  
int main()  
{  
    char s[100] = "abcdefg";  
    char t[100];  
    strcpy(t,s);  
    reverse_string(s);
```

```
if(strcmp(t,s) == 0)
    printf("Yes");
else
    printf("No");
return 0;
}
```

- a) Copies a string to another string
- b) Compares two strings
- c) Reverses a string
- d) Checks if a string is a palindrome

Ans. D

45. Consider the following iterative code used to convert a decimal number to its equivalent binary:

```
#include<stdio.h>
void dec_to_bin(int n)
{
    int arr[31],len = 0,i;
    if(n == 0)
    {
        arr[0] = 0;
        len = 1;
    }
    while(n != 0)
    {
        arr[len++] = n % 2;
        _____;
    }
    for(i=len-1; i>=0; i--)
        printf("%d",arr[i]);
}
int main()
```



```
{  
    int n = 10;  
    dec_to_bin(n);  
    return 0;  
}
```

Which of the following lines should be inserted to complete the above code?

- a) n-
- b) n /= 2
- c) n /= 10
- d) n++

Ans. B

46. What is the output of the following code?

```
#include<stdio.h>  
int arr[31], len = 0;  
void recursive_dec_to_bin(int n)  
{  
    if(n == 0 && len == 0)  
    {  
        arr[len++] = 0;  
        return;  
    }  
    if(n == 0)  
        return;  
    arr[len++] = n % 2;  
    recursive_dec_to_bin(n/2);  
}  
int main()  
{  
    int n = -100, i;  
    recursive_dec_to_bin(n);  
    for(i=len-1; i>=0; i--)
```

```
    printf("%d",arr[i]);  
    return 0;  
}
```

- a) -1100100
- b) 1100100
- c) 2's complement of 1100100
- d) Garbage value

Ans. D

47. What is the output of the following code?

```
#include<stdio.h>  
int get_len(char *s)  
{  
    int len = 0;  
    while(s[len] != '\0')  
        len++;  
    return len;  
}  
int main()  
{  
    char *s = "";  
    int len = get_len(s);  
    printf("%d",len);  
    return 0;  
}
```

- a) 0
- b) 1
- c) Runtime error
- d) Garbage value

Ans. A

48. Which of the following correctly represents the function to insert elements at the bottom of stack?

a)

```
int BottomInsert(int x)
{
    if(s.size()!=0) s.push(x);
    else
    {
        int a = s.top();
        s.pop();
        BottomInsert(x);
        s.push(a);
    }
}
```

b)

```
int BottomInsert(int x)
{
    if(s.size()==0) s.push(x);
    else
    {
        int a = s.top();
        s.pop();
        s.push(a);
        BottomInsert(x);
    }
}
```

c)

```
int BottomInsert(int x)
{
    if(s.size()==0) s.push(x);
    else
    {
```

```
        int a = s.top();
        s.pop();
        BottomInsert(x);
        s.push(a);
    }
}
```

d)

```
int BottomInsert(int x)
{
    if(s.size()==0) s.push(x);
    else
    {
        s.pop();
        int a = s.top();
        BottomInsert(x);
        s.push(a);
    }
}
```

Ans. C

49. Consider the following code snippet to find the smallest element in an array:

```
int get_min_element(int *arr, int n)
{
    int i, min_element = arr[0];
    for(i = 1; i < n; i++)
        if(_____)
            min_element = arr[i];
    return min_element;
}
```

Which of the following lines should be inserted to complete the above code?

a) `arr[i] > min_element`

- b) arr[i] < min_element
- c) arr[i] == min_element
- d) arr[i] != min_element

Ans. B

50.Consider the following code snippet to find the largest element in a linked list:

```
struct Node{
    int val;
    struct Node *next;
}*head;
int get_max()
{
    struct Node* temp = head->next;
    int max_num = temp->val;
    while(_____)
    {
        if(temp->val > max_num)
            max_num = temp->val;
        temp = temp->next;
    }
    return max_num;
}
```

Which of the following lines should be inserted to complete the above code?

- a) temp->next != 0
- b) temp != 0
- c) head->next != 0
- d) head != 0

Ans. b